A Contact-Centred Benchmark for Soft Passive-Adaptive vs Rigid Parallel-Jaw Grippers on Thin, Flat Cloth

Shiyao Ni
McGill University
shiyao.ni@mail.mcgill.ca

Abstract

Robotic manipulation of thin, flat cloth resting on a hard surface remains an open problem. Most existing solutions augment rigid parallel-jaw grippers with environment-assisted actions, specialised end-effectors, or closed-loop perception, treating the gripper itself as a fixed component. Prior benchmarks for cloth manipulation either standardise tasks across many gripper designs without instrumenting the contact interaction [1], or compare soft and rigid grippers under a single fixed-condition test with an actively-controlled pneumatic soft hand [2]. This proposal targets the contact stage directly: a benchmark that explicitly treats gripper compliance as the independent variable and maps the per-gripper success envelope in a five-axis control-parameter space, comparing a soft, passively-adaptive UMI finray (TPU 90A) against a factory Franka parallel jaw on the same Franka Research 3 arm under an identical control policy. The output is a pair of envelopes whose shape and area form the basis of the soft-vs-rigid comparison, complemented by three functional cloth-manipulation tasks under a shared human-defined policy.

1 Problem

Current robot-manipulation research relies heavily on the operator to hand the target object into the manipulator before any trajectory is executed. This work targets the upstream step that is typically skipped: the robot picking up a thin, flat cloth lying on a hard surface, without human staging and without active replanning. The specific failure mode under study is the contact stage of the grasp, where the gripper must make controlled contact with the cloth–surface system and pinch a graspable layer.

2 Motivation

If a robot cannot pick up cloth flat from a surface, the practical autonomy of downstream manipulation pipelines is severely constrained. Successful flat-cloth grasping enables:

- Laundry folding and textile handling;
- Hospital bedding management;
- Assistive robotics for elderly or disabled individuals;
- Garment manufacturing and logistics;
- Soft-object packaging.

Each application sits behind the same gating capability: lifting a single flat layer from a hard surface reliably.

3 State of the Art

Current solutions combine environment interaction, specialised end-effectors, and feedback mechanisms to make cloth grasping tractable.

Environment-assisted actions create graspable features (folds, wrinkles, exposed edges) through dragging, wiping, or peeling, allowing a standard parallel-jaw gripper to perform a pinch on a prepared edge.

Specialised end-effectors reduce reliance on precise edge access: variable-friction grippers, micro-needle or needle-based graspers, electroadhesion, and airflow-based methods (Bernoulli or Coandă effects).

Closed-loop perception and control verify grasp success (e.g., single-ply detection) and trigger recovery actions when failures occur. In practice, most systems still rely on rigid parallel-jaw grippers augmented with these strategies.

Two recent works are directly relevant to this proposal because they attempt cross-gripper comparison rather than perception or planning improvements. Clark *et al.* [1] introduce a standardised household-clothing object set and four benchmark tasks (edge drag accuracy, edge grasp resilience, crumpled object encapsulation, flat non-boundary grasp success), and evaluate four parallel grippers including a Fin Ray® variant. However, their benchmarks score outcomes (placement offset, payload, droop, binary success) rather than instrument the contact stage; the soft-vs-rigid difference is reported but not isolated from the task-specific gripper kinematics.

Teeple *et al.* [2] directly compare a rigid parallel jaw to a soft pneumatic two-finger gripper, showing that vertical, lateral, and rotational compliance each protect against distinct failure modes (positional uncertainty, snag duration, peak snag force). Their soft gripper, however, is actively pressure-controlled, and the comparison is performed at a fixed vertical sweep without a multi-axis control-parameter envelope.

4 Drawbacks of the State of the Art

- Heavy reliance on environment-assisted actions reduces true autonomy and inflates task complexity and execution time.
- Specialised end-effectors are task-specific, hard to generalise, and often fragile or impractical for real-world deployment.
- Rigid parallel-jaw grippers require precise positioning and well-defined geometry, making them unreliable for thin, flat cloth on a hard surface.
- Closed-loop verification and recovery introduce sensing and control overhead.
- Most approaches treat the gripper as fixed and focus on perception or planning, so there is no systematic evaluation of how gripper design (rigid vs. soft) affects the contact stage in isolation.

Consequently, directly grasping thin, flat cloth from a surface remains an open problem. The work that does compare soft and rigid grippers either does not instrument the contact [1] or relies on active control of the soft side [2], leaving the contribution of a *passive* soft gripper at the contact stage uncharacterised.

5 Contribution

This proposal contributes a benchmark focused on the contact stage of grasping thin, flat cloth, comparing a soft but passively-adaptive gripper (UMI finray, TPU 90A; no added control space relative to a parallel jaw) against a rigid parallel-jaw gripper on the same robot arm under the same human-defined control policy.

5.1 Benchmark Metrics

Control-policy admissibility: a trial is admissible only if the Franka controller did not quit due to a force-safety error during execution. Per-trial outcome is determined by a researcher visual verification of grasp success, combined with the controller-status flag.

The benchmark sweeps five gripper-control axes (approach z-offset, head rotation, applied-force setpoint, finger gap, closing speed) and records the boundary of the region in which both conditions hold. Five tolerance metrics per gripper are then read off the envelope:

1. **z-axis tolerance:** range of approach-height offset over which the grasp succeeds.
2. **Angular tolerance:** orientation spread of the Franka achieved end-effector pose observed across successful trials with commanded orientation held constant (a passive observation, not a swept parameter).
3. **Maximum admissible surface-contact force** at which the controller still agrees (does not quit).
4. **Peak contact force** during successful grasps (table or cloth side).
5. **Minimum contact force** required for a successful grasp.

5.2 Functional Tasks

Both grippers are additionally tested on three downstream tasks under a single shared human-defined policy:
- Diagonal fold (flat cloth corner grasp);
- Middle pick → drop onto adjacent sphere (flat cloth middle grasp);
- Drag from A to B (flat cloth edge grasp).

A future scope extension may add tasks dealing with folded cloth.

5.3 Comparison with Prior Benchmarks

Table 1 situates this work alongside the two most directly comparable prior benchmarks. Conceptually, the proposal sits closer to Teeple *et al.* [2] (same soft-vs-rigid question on flat thin cloth) than to Clark *et al.* [1] (broad standardised benchmark across many gripper designs), but it replaces Teeple's two-point comparison with a five-axis swept envelope and instruments the under-cloth force-distribution side with a distributed force-sensor array rather than a single wrist force/torque sensor.

Distinctive contributions of this work relative to both prior benchmarks:
1. *Envelope shape, not aggregate pass-rate.* The output is a region in 5-D control-parameter space whose boundary is the metric, not a per-task scalar score.
2. *Controller-quit as a tracked admissibility gate.* Neither prior benchmark explicitly separates "controller refused to continue" from "grasp failed." Teeple's rigid-gripper failures include UR5e safety-stop trips that are conflated with grasp failure. This benchmark separates them to avoid biasing the comparison against grippers operating near the controller's force-safety threshold.
3. *Distributed under-cloth force sensing.* A five-channel force array (Tekscan A301-1) measures per-foot table-side reaction force; a two-channel A301-25 pair measures finger-pad pinch force; the Franka built-in F_{ext} and joint torques are logged in parallel.
4. *Force traceability.* Per-channel calibration is traceable to an OIML M1 mass standard via dead-weight loading with a power-law fit; calibration freshness is a hard gate on data admissibility.
5. *Passive soft gripper.* The soft side is a passively deforming UMI finray, not an active pneumatic system, isolating the contribution of compliance to grasp success without introducing additional control variables.

6 How the Contribution Addresses the Drawbacks

- **Gripper as a variable.** By explicitly treating the gripper as the independent variable rather than a fixed subsystem, this work addresses the central limitation identified in Section 4: that gripper design is rarely the swept axis in cloth-manipulation benchmarks.
- **Quantified behaviour under uncertainty.** The proposed metrics (positional tolerance, angular tolerance, force profiles) directly quantify how each gripper behaves under control-parameter uncertainty. This reduces reliance on environment-assisted actions such as dragging or folding to expose graspable features.
- **Isolated compliance effect.** By comparing a soft passively-adaptive gripper to a rigid parallel jaw under the same control policy, the study isolates the contribution of compliance and contact mechanics, rather than conflating it with perception or planning improvements. This is the gap left open by [1] (does not instrument contact) and [2] (active pneumatic control on the soft side).
- **Force-related failure modes.** The minimum, peak, and maximum-admissible force metrics provide visibility into slipping, snagging, and excessive-force failure modes that are not surfaced by current benchmarks.
- **Task-level usefulness.** The three functional tasks ensure that grasps are not only successful at contact but also useful for downstream manipulation, addressing the gap between grasp acquisition and task execution.
- **Hardware-aware design.** The overall framing shifts part of the cloth-manipulation problem from complex perception and control toward hardware-aware design, potentially reducing system complexity while improving robustness on thin, deformable objects.

Table 1: Comparison of this proposal with two directly relevant prior benchmarks for cloth grasping.

Axis	**This proposal**	**Clark et al. 2023 [1]**	**Teeple et al. 2022 [2]**
Research question	Map the control-parameter success envelope per gripper (size and shape of the region in which the grasp succeeds and the Franka controller does not quit).	Define a standardised cloth-manipulation benchmark suite (object set + edge taxonomy + four tasks) for cross-paper gripper comparison.	Show that vertical, lateral, and rotational compliance each protect against distinct failure modes (uncertainty, snag duration, peak snag force).
Output type	Two parameter-space envelopes, one per gripper, compared in shape and area.	Per-gripper score on each of four tasks (offset mm, payload N, droop mm, success %).	Range of vertical-offset tolerance (mm); snag displacement (mm, deg), force duration (s), peak lateral force (N).
Object set	2 cloth specimens (1 thin + 1 thick), flat only.	13-item household clothing set with edge classification (single/double/folded; boundary/non-boundary).	3 swatches: elastic spandex, woven cotton, folded woven cotton.
Cloth state(s) tested	Flat only (crumpled/folded out of scope).	Flat boundary edges, non-boundary edges, crumpled, compressed-crumpled.	Flat only; snags imposed by external clamp.
Grippers compared	22 mm UMI finray (TPU 90A), 17 mm UMI finray (TPU 90A), Franka factory parallel jaw (**3**).	OpenHand T42, Robotiq 2F-140, FE Hand, FE Hand (Fin Ray Flex) (**4**).	Rigid PhantomX AX-12 (foam pads) vs custom soft pneumatic 2-finger (**2**).
Independent variable	*Swept:* approach z-offset, head rotation, applied force, finger gap, closing speed.	*Categorical:* object × edge type × gripper.	*Swept:* vertical centring offset (-40 to $+4$ mm); other parameters fixed.
Robot	Franka Research 3.	Franka Emika Panda.	UR5e (150 N safety stop).
Surface / fixturing	Instrumented 3D-printed test bench; rigid feet on A301-1 array (TPU 95A 1 mm contact pad option per Bill 0003 amendment).	A0 alignment grid on rigid backing plate.	Plain tabletop.
Force instrumentation	5× A301-1 table feet + 2× A301-25 finger pads + Franka F_{ext} + Franka joint torques, shared ROS timebase; 100 Hz DAQ + 1 kHz Franka FCI.	Single load cell on motorised rail (resilience benchmark only); 1 mm ruler for placement.	UR5e wrist F/T sensor (500 Hz) + AprilTag pose tracking (30 Hz).
Force calibration rigor	OIML M1 traceable dead-weight per-channel power-law fit; JSON record with derivation header; 30-day re-cal cadence; σ-discard threshold.	Not specified.	Stiffness measured (n=3); per-trial force traceability not specified.
Pass/fail criterion	Researcher visual verification + Franka controller-not-quit admissibility gate (treated as separate failure class).	Binary success per task.	Binary “arm picks up swatch.”
Distinctive failure modes tracked	Controller-quit / force-safety abort tracked as a signal-derived admissibility class.	Drop during motion; lift failure; encapsulation failure.	UR5e safety-stop and grasp instability (conflated with grasp failure for the rigid gripper); tear under snag.
Trials per condition	Parameter sweep (count per axis TBD at execution).	5 repeats per object.	n=3 per condition.

7 Grasping Strategy

The proposed contact protocol is a multi-stage descent followed by a slow close and slow ascent. Defining the cloth–surface contact plane as $z = 0$:

1. **Multi-stage descent.** Descend rapidly to the (x, y) contact position at $z = +5$ mm. Slow descent to $z = 0$ mm. Very slow descent to a small negative height (e.g., $z = -3$ mm). This provides a safety margin and a force limit, and allows the finray to deform slightly. The resulting contact force between the lower section of the gripper and the cloth enables the subsequent steps.
2. **Slow close.** Close the gripper slowly. Because contact force already exists between the lower section

of the gripper and the cloth, this closing motion pinches the cloth into the inner contact surface of the gripper.

3. **Slow ascent.** Ascend slowly in the first few steps to allow the finray to recover its original shape and grip the cloth firmly.
4. **Pinch margin.** The margin between the gripper's current opening and the cloth thickness determines the amount of fabric pinched into the contact zone.
5. **Comparison with parallel jaw.** The protocol is enabled by the deformable nature of the finray. The Franka parallel jaw can succeed on the same task but does not require (and cannot exploit) the negative-height step. Without that step, the parallel jaw requires careful height tuning: too-high fails on insufficient contact; too-low fails on excessive contact force triggering the controller-quit admissibility gate. Quantifying this tuning margin per gripper is precisely the z-axis tolerance metric in Section 5.

References

[1] A. B. Clark, L. Cramphorn-Neal, M. Rachowiecki, and A. Gregg-Smith, "Household clothing set and benchmarks for characterising end-effector cloth manipulation," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2023.

[2] C. B. Teeple, J. Werfel, and R. J. Wood, "Multi-dimensional compliance of soft grippers enables gentle interaction with thin, flexible objects," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2022, pp. 728–734.